Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/04/2007
PI Name:	Soller, Babs R. Ph.D.		
Project Title:	Noninvasive Measurement of Blood and Tissue Chemistry: SMS00205		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Smart Medical Systems Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	babs.soller@reflectancemedical.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	508-366-4700
Organization Name:	Reflectance Medical Inc.		
PI Address 1:	116 Flanders Road, Suite 1000		
PI Address 2:			
PI Web Page:			
City:	Westboro	**State:**	MA
Zip Code:	01581	**Congressional District:**	3
Comments:	Previous affiliation was University of Massachusetts Medical School, Worcester, MA, until October 2010 (Ed., 11-9-2010)		
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	03/01/2001	**End Date:**	03/31/2004
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Heard, Stephen (University of Massachusetts Med School) Puyana, Juan (University of Pittsburgh)		
Grant/Contract No.:	NCC 9-58-SMS00205		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The long term goal of this project is to create a noninvasive and lightweight system for measuring blood and tissue chemistry through the skin. The technology is based on Near Infrared Spectroscopy (NIRS) and is targeted to work on subjects of any ethnic origin. The system developed during this 3 year project will measure three metabolic parameters: muscle pH, muscle PO2 (oxygen tension) and blood hematocrit. The system can be part of smart medical systems used to identify and treat traumatic injury and assess fitness levels on earth and in space. It can also assist in the development of more time efficient exercise countermeasures. We developed novel algorithms to simultaneously and noninvasively measure muscle pH, PO2 and blood hematocrit. We demonstrated human feasibility of this measurement on 18 patients undergoing cardiac surgery. The measurement algorithms applied to individual patients were accurate enough to detect small changes in local peripheral perfusion resulting from reduced blood pressure and temperature-induced changes in metabolic demand, despite significant variation in body temperature. Physiological changes during cardiac surgery model mild shock. We expect seriously injured patients to have significantly larger variation in the noninvasively measured parameters. We have also developed novel techniques to correct transdermally collected spectra for the interfering effect of skin color. The technique was shown to improve the accuracy of hemoglobin measurement on tissue-mimicking phantoms. We also have shown that we can derive the skin color correction factor from human subject spectra. We found that the spectroscopic equipment and probe placement techniques used for the cardiac surgery study were not stable and precise enough to allow calibration equations to be developed across multiple subjects. With funding from the US Army Medical Research Command, we have designed and built new hardware to overcome these difficulties. We collected a new set of calibration data from cardiac surgery patients with this equipment and were able to develop multi-subject calibration models with accuracy near our target values. However, the range of pH and PO2 values for these calibration models was limited compared to our original set of patient data and too small for use in an exercise study. We have developed a new human subject protocol using hand dynamometry and will use data from this study do develop calibration equations which have the range required not only for exercise studies, but serious traumatic injury. We developed two layer phantoms to spectroscopically study fat layers over muscle. We also developed computer-based Monte Carlo simulation models to study how variability in optical coefficients for muscle and fat would impact human spectra. Input from the models helped in the design of a modified fiber optic sensor to measure muscle spectra beneath a thick fat layer. We also modified our optical hardware to achieve good spectral signal from muscle beneath thick fat layers. The sensor and monitor have been successfully tested on phantoms with thick fat layers. Monte Carlo simulation showed that for most subjects, no fat corrections would be needed for NIRS measurements made on the forearm. We also used Monte Carlo simulations and phantom measurements to evaluate potential fat correction algorithms. Work in the next year will focus on implementation of the most promising approaches for fat correction, culminating in a demonstration on human subjects. NIRS-based noninvasive measurement of muscle pH, PO2 and blood hematocrit has been demonstrated on human subjects and techniques are being implemented to ensure that these measurements are accurate for all subjects, irrespective of fat thickness or skin color. We expect application in the treatment of critically ill subjects and for assessing exercise intensity and muscle fatigue. Next year we anticipate beginning validation of the sensor using specific NASA exercise protocols. We will also begin a study of Emergency Room patients in shock to derive values for muscle pH and PO2 which can be used to guide resuscitation through a smart medical system.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will have direct earth based application. The prototype monitors we are developing will have application in emergency response vehicles, emergency rooms, and hospitals. Pre-hospital applications include assessing the severity of shock and triaging multiple casualties, as well as providing a sensor for a smart medical system to guide resuscitation from hemorrhage. In the ICU we expect that this monitor will find application in distinguishing between hemorrhagic and septic shock. The direct muscle application of interest to NASA for assessing fitness in space, may be useful to assess success of physical therapy in rehabilitating patients with muscle injury or atrophy. There is also general medical application for the diagnosis of anemia and if small and inexpensive enough, screening world-wide for malnutrition. There is also a possible application for the diagnosis of diabetic foot ulcers. A smaller version of this monitor could find use in the training of elite and weekend athletes.
Task Progress:	SPECTRAL CORRECTIONS FOR SKIN COLOR AND FAT THICKNESS We invented a new, easier approach for skin color correction. Instead of collecting spectra on 100 subjects with variation in skin color, we vary the skin color of a single individual with artificial coloring. This was demonstrated on two-layer solid phantoms with melanin in the top layer and hemoglobin in the bottom layer. The new method not only visually removed the absorption of melanin from the spectra, but also improved the accuracy of the hemoglobin measurement. We have developed methods to artificially color human skin and extracted a skin color factor that is nearly identical to that derived from the melanin phantoms. We also developed two layer phantoms to evaluate approaches to correct for the optical interference from fat. Our initial attempt resulted in a new method to noninvasively measure fat thickness, but did not effectively correct the spectra for use in further calculations. A new approach employing a sensor design with 2 different source-detector spacings and novel mathematics is under evaluation. DEVELOP CALIBRATION EQUATIONS TO MEASURE PH, PO2 AND HEMATOCRIT ON HUMAN SUBJECTS Work continued on multi-subject calibration models using the data from the first 18 cardiac surgery patients, but was unsuccessful due to equipment limitations and lack of precision in placing the sensor. A small, portable spectroscopic monitor was designed and built with the assistance of Luxtec Corporation and related funding from the US Army Medical Research Command. The new monitor contains novel circuit and instrument design to improve stability. Additionally, we developed a targeting system to improve placement precision. Multi-subject calibration models for muscle pH, PO2 and blood hematocrit were calculated from a new set of surgery patients and had accuracy near our target values. The range of pH and PO2 values for these models was limited, compared to the original set, and too small for use in an exercise study. We defined a new human subject protocol using hand grip dynamometry and will use data from this study do develop calibration equations which have the range required for both exercise studies and trauma patients. SPECIFY REQUIREMENTS FOR MINIATURE, LIGHTWEIGHT OPTICAL DEVICE We developed specifications for a small, lightweight portable near infrared spectrometer. With related funding from the US Army Medical Research Command and the help of Luxtec Corporation we built 3 prototype of this small system. The system weighs 8 lbs and is approximately 8”W x 9”L x 5”D.

	DESIGN OF THE OPTICAL SYSTEM FOR MEASUREMENT ON THE LEG We used 2-layer phantoms and Monte-Carlo computer simulations to determine specifications for a sensor and monitor to be used to assess muscle metabolism on the leg, through a thick layer of fat. The system has been constructed is currently being used to assess fat correction techniques.
Bibliography Type:	Description: (Last Updated: 03/22/2019)
Articles in Peer-reviewed Journals	Soyemi OO, Landry MR, Yang Y, Idwasi PO, Soller BR. "Skin color correction for tissue spectroscopy: demonstration of a novel approach with tissue-mimicking phantoms." Appl Spectrosc. 2005 Feb;59(2):237-44. PMID: 15720765 , Feb-2005
Patents	60/564,955. Provisional patent application, 2004. Jan-2004 Landry M.R, Soller BR, Previte GJ. "Alignment Target for Precise Probe Placement."
Patents	60/565,655. Provisional patent application, 2004. Jan-2004 Phillipps P, Phillipps, M, Soller BR. "Stabilized Filament Light Source. "
Patents	10/269,826. Patent, May 2004. May-2004 Soller BR. "Noninvasive Measurement of Tissue Oxygenation."
Patents	10/086,917. Patent, May 2004. May-2004 Soller, B. R., and I. P. Idwasi. "Correction of Spectra for Subject Diversity."
Patents	60/565,557. Provisional patent application, Jan 2004. Jan-2004 Soller BR, Phillipps P, Phillipps M, Soyemi O, Parker M, DeMilo C, Yonkers EH, Perry W. "Spectrosopic System for Optical Reflectance Measurements. "